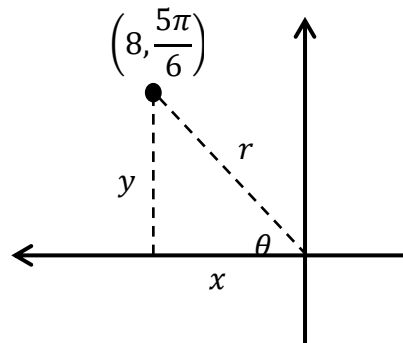


5.2 Polar Worksheet

1. A polar point has the coordinates. Find and fully simplify the Cartesian coordinates of this point.

$$x = 8 \cos\left(\frac{5\pi}{6}\right) = 8\left(-\frac{\sqrt{3}}{2}\right) = -4\sqrt{3}$$

$$y = 8 \sin\left(\frac{5\pi}{6}\right) = 8\left(\frac{1}{2}\right) = 4$$



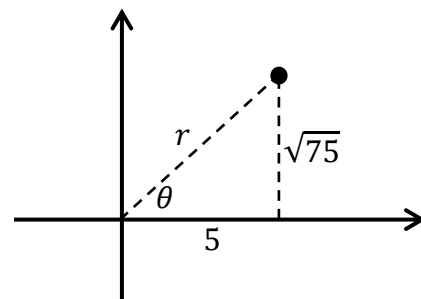
2. A Cartesian point has the coordinates $(5, \sqrt{75})$. Find a polar coordinate representation for this point that has a positive value for r and another representation that has a negative value for r .

$$r^2 = x^2 + y^2 = (5)^2 + (\sqrt{75})^2 = 100 \Leftrightarrow r = \pm\sqrt{10}$$

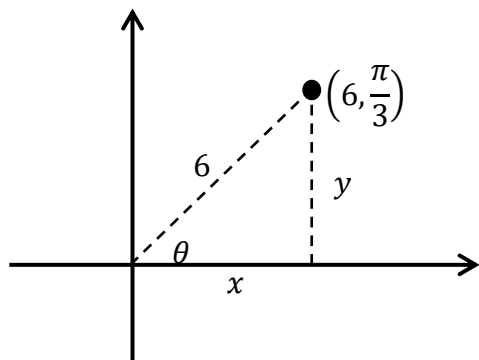
$$\tan(\theta) = \frac{\sqrt{75}}{5} = \frac{5\sqrt{3}}{5} = \sqrt{3} \Leftrightarrow \theta = \frac{\pi}{3} + \pi k \text{ for some integer } k.$$

Therefore we can create the possible following polar representations of the point (*Note: there are infinitely many representations*)

$$\left(10, \frac{\pi}{3}\right) \text{ or } \left(-10, \frac{4\pi}{3}\right)$$



3. Convert $\left(6, \frac{\pi}{3}\right)$ to rectangular coordinates.

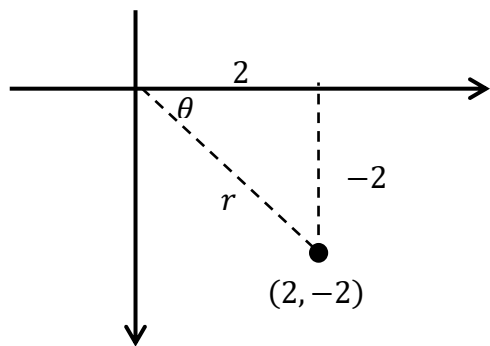


$$x = 6 \cos\left(\frac{\pi}{3}\right) = 6\left(\frac{1}{2}\right) = 3$$

$$y = 6 \sin\left(\frac{\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$$

The rectangular coordinates of the point are: $(3, 3\sqrt{3})$

4. Convert $(2, -2)$ to polar coordinates.



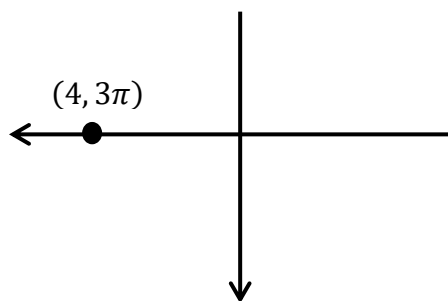
$$r = \sqrt{(2)^2 + (-2)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$

$$\sin(\theta) = -\frac{\sqrt{2}}{2} \rightarrow \theta = -45^\circ = -\frac{\pi}{4}$$

Some possible polar coordinates of the point are:
 $(2\sqrt{2}, -45^\circ)$ or $(2\sqrt{2}, 315^\circ)$ or $(-2\sqrt{2}, 135^\circ)$

5. Convert $(4, 3\pi)$ to rectangular coordinates.

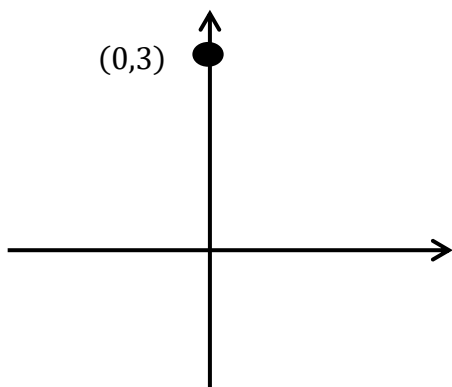
This point lies on an axis, so the conversion is simple.



$$x = -4 \text{ \& } y = 0$$

The rectangular coordinates of the point are: $(-4, 0)$

6. Convert $(0, 3)$ to at least 3 different points in polar coordinates.



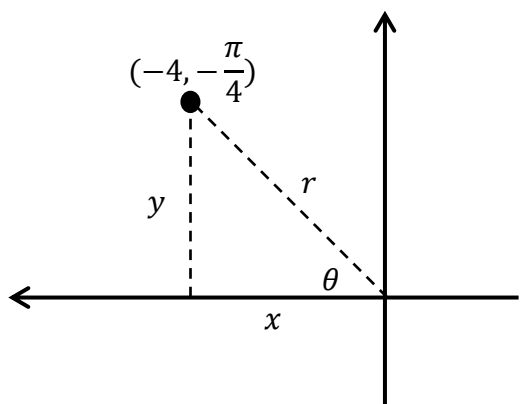
This point lies on an axis, so the conversion is simple.

$$r = 3 \text{ \& } \theta = \frac{\pi}{2}$$

Some possible polar coordinates of the point are:

$$\left(3, \frac{\pi}{2}\right) \text{ or } \left(3, \frac{5\pi}{2}\right) \text{ or } \left(-3, \frac{3\pi}{2}\right)$$

7. Convert $\left(-4, -\frac{\pi}{4}\right)$ to rectangular coordinates.

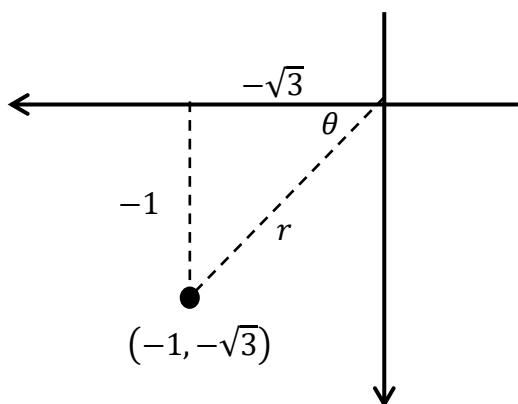


$$x = -4 \cos\left(-\frac{\pi}{4}\right) = -4 \left(\frac{\sqrt{2}}{2}\right) = -2\sqrt{2}$$

$$y = -4 \sin\left(-\frac{\pi}{4}\right) = -4 \left(-\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$$

The rectangular coordinates of the point are: $(-2\sqrt{2}, 2\sqrt{2})$

8. Convert $(-\sqrt{3}, -1)$ to polar coordinates.



$$r = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2$$

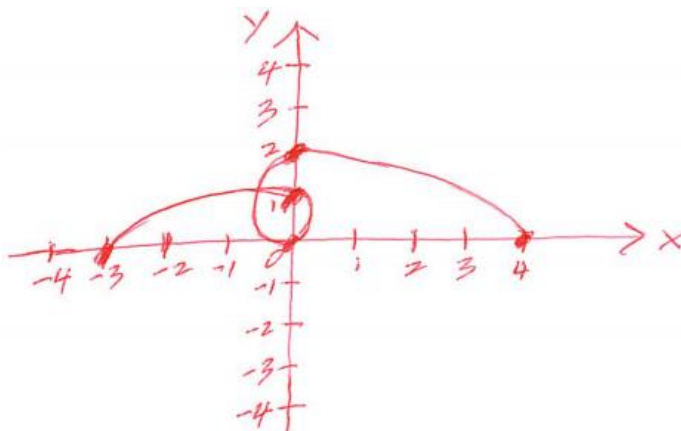
$$\sin(\theta) = -\frac{1}{2} \rightarrow \theta = 180^\circ - \sin^{-1}\left(-\frac{1}{2}\right) = 180^\circ + 30^\circ = 210^\circ$$

Note that $\theta \neq \sin^{-1}\left(-\frac{1}{2}\right)$ since θ is in quadrant III and $\sin^{-1}\left(-\frac{1}{2}\right)$ must return an angle in quadrant IV.

Some possible polar coordinates of the point are:
 $(2, 210^\circ)$ or $(2, -150^\circ)$ or $(-2, 30^\circ)$

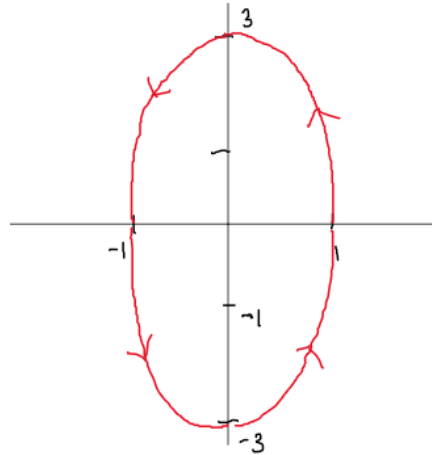
9. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r decreases from 4 to 2.
- As θ increases from $\pi/2$ to π , r decreases from 2 to 0.
- As θ increases from π to $3\pi/2$, r decreases from 0 to -1.
- As θ increases from $3\pi/2$ to 2π , r decreases from -1 to -3.



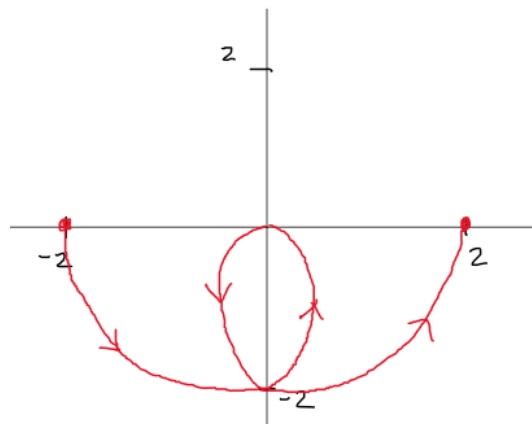
10. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r increases from 1 to 3.
- As θ increases from $\pi/2$ to π , r decreases from 3 to 1.
- As θ increases from π to $3\pi/2$, r increases from 1 to 3.
- As θ increases from $3\pi/2$ to 2π , r decreases from 3 to 1.



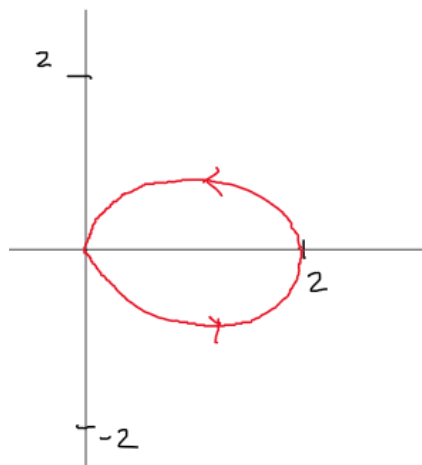
11. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r stays constant at -2.
- As θ increases from $\pi/2$ to π , r increases from -2 to 0.
- As θ increases from π to $3\pi/2$, r increases from 0 to 2.
- As θ increases from $3\pi/2$ to 2π , r stays constant at 2.



12. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

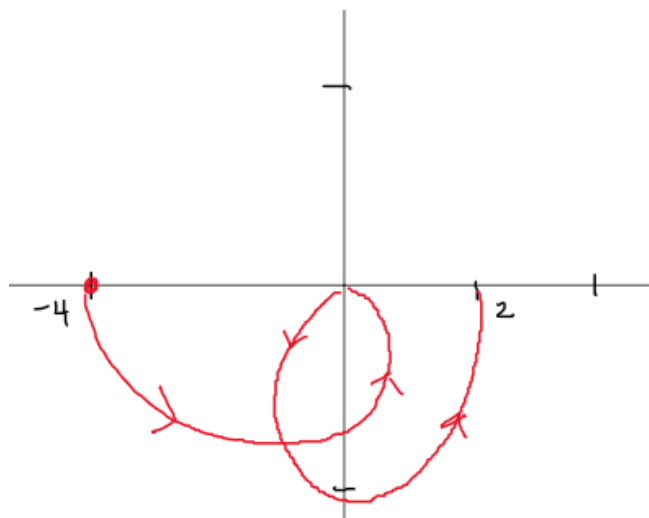
- As θ increases from 0 to $\pi/2$, r decreases from 2 to 0.
- As θ increases from $\pi/2$ to π , r decreases from 0 to -2.
- As θ increases from π to $3\pi/2$, r increases from -2 to 0.
- As θ increases from $3\pi/2$ to 2π , r increases from 0 to 2.



Note that the 3rd and 4th bullet points above could just create a curve that overlaps the curves created by the first two bullet points. This is what is pictured in the image above.

13. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- At $\theta = 0$, the curve is at $(-4, 0)$.
- Between $0 < \theta < \pi$, r is always negative and increasing.
- Between $\pi < \theta < 3\pi/2$, r is always positive and increasing.
- Between $3\pi/2 < \theta < 2\pi$, r is always positive and decreasing.
- $r = 0$ only when $\theta = \pi$.
- At $\theta = 2\pi$, the curve is at $(2, 0)$



14. Convert the polar equation $r = \frac{1}{5 \cos(\theta) + 3 \sin(\theta)}$ to a Cartesian equation and solve this resulting equation for y .

$$r = \frac{1}{5 \cos(\theta) + 3 \sin(\theta)} \Leftrightarrow 5r \cos(\theta) + 3r \sin(\theta) = 1 \Leftrightarrow 5x + 3y = 1 \Leftrightarrow y = -\frac{5}{3}x + \frac{1}{3}$$

15. Convert the following polar equation to Cartesian coordinates $r = 2 \sec(\theta)$.

$$r = 2 \sec(\theta) \Rightarrow r = \frac{2}{\cos(\theta)} \Rightarrow r \cos(\theta) = 2 \Rightarrow x = 2$$

16. Convert the following rectangular equation to polar coordinates $x = \frac{3}{y}$.

$$x = \frac{3}{y} \Rightarrow r \cos(\theta) = \frac{3}{r \sin(\theta)} \Rightarrow r^2 \sin(\theta) \cos(\theta) = 3$$

17. Convert the following polar equation to Cartesian coordinates $r \sin(\theta) = r \sqrt{\cos(2\theta)}$.

$$\begin{aligned} r \sin(\theta) &= r \sqrt{\cos(2\theta)} \Rightarrow y = r \sqrt{\cos(2\theta)} \Rightarrow y^2 = r^2 \cos(2\theta) \Rightarrow y^2 = r^2 [\cos^2(\theta) - \sin^2(\theta)] \\ &\Rightarrow y^2 = r^2 \cos^2(\theta) - r^2 \sin^2(\theta) \\ &\Rightarrow y^2 = x^2 - y^2 \\ &\Rightarrow 2y^2 - x^2 = 0 \end{aligned}$$

18. Convert the following rectangular equation to polar coordinates $y^2 = 5x$.

$$\begin{aligned} y^2 &= 5x \Rightarrow [r \sin(\theta)]^2 = 5r \cos(\theta) \Rightarrow r^2 \sin^2(\theta) = 5r \cos(\theta) \\ &\Rightarrow r = 5 \cot(\theta) \csc(\theta) \end{aligned}$$